

CSU – RAMS

Comprehensive Analysis File Variable List

This document contains the full list of output variables as they appear in the native HDF5 output analysis files. Included are the name, units, and description of the variables. Note that not all of the listed variables are available from every simulation. Variables are not output unless the physics schemes they are associated with are turned on in the simulations via user choices in the RAMSIN namelist.

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Last updated: 31 August 2026 (for v6.3.05)

RAMS Variables (prior to REVU post-processing)			
ASCII Name	dimensions	units	Description
GRID AND TOPOGRAPHY			
GLAT	nx,ny	deg	Latitude
GLON	nx,ny	deg	Longitude
TOPT	nx,ny	m	topography height
TOPZO	nx,ny	m	topographic roughness length

DYNAMICS & THERMODYNAMICS			
PI	nx,ny,nz	J/(kg*K)	PI = Exner function * Cp, where (Cp=1004 J/kg/K in RAMS) Exner-function = $T/\theta = (p/p00)^{(Rd/Cp)}$
PIO	nx,ny,nz	J/(kg*K)	(INPUT) reference state PI
UP	nx,ny,nz	m/s	Past U (zonal) wind component
VP	nx,ny,nz	m/s	Past V (meridional) wind component
WP	nx,ny,nz	m/s	Past W (vertical) wind component
PP	nx,ny,nz	J/(kg*K)	Past perturbation Exner function (PI-prime)
UC	nx,ny,nz	m/s	Current U wind component
VC	nx,ny,nz	m/s	Current V wind component
WC	nx,ny,nz	m/s	Current W wind component
PC	nx,ny,nz	J/(kg*K)	Current perturbation Exner function (PI-prime)
THP	nx,ny,nz	K	Theta-IL, ice-liquid potential temperature (prognostic variable)
RTP	nx,ny,nz	kg/kg	Total water mixing ratio (water vapor + condensate) (prognostic variable)
THETA	nx,ny,nz	K	Theta, potential temperature
RV	nx,ny,nz	kg/kg	Water vapor mixing ratio
DN0	nx,ny,nz	kg/m³	reference state air density

SURFACE FILE INPUT CHARACTERISTICS (Used by both LEAF & SIB)			
(LEAF always runs PATCH=1, which is water/ocean)			
SOIL_TEXT	nx,ny,nzg,np	#	dominant soil textural class
LEAF_CLASS	nx,ny,np	#	vegetation class
PATCH_AREA	nx,ny,np	fraction	patch fractional area

VEG_NDVIP	nx,ny,np	#	past NDVI (NDVI = Normalized Difference Vegetation Index)
VEG_NDVIC	nx,ny,np	#	current NDVI
VEG_NDVIF	nx,ny,np	#	future NDVI

SURFACE CHARACTERISTICS (LEAF3 / SIB)
(Diagnostic for both)

PATCH_ROUGH	nx,ny,np	m	net roughness
SOIL_ROUGH	nx,ny,np	m	soil roughness
VEG_FRACAREA	nx,ny,np	fraction	vegetation fractional area
VEG_LAI	nx,ny,np	m ² /m ²	green leaf area index
VEG_TAI	nx,ny,np	m ² /m ²	total leaf area index
VEG_ROUGH	nx,ny,np	m	vegetation roughness length
VEG_HEIGHT	nx,ny,np	m	vegetation / canopy height
VEG_ALBEDO	nx,ny,np	fraction	vegetation albedo

SURFACE CHARACTERISTICS (LEAF3 / SIB)
(Prognostic for both)

SOIL_WATER	nx,ny,nzg,np	m ³ /m ³	volumetric soil moisture
SOIL_ENERGY	nx,ny,nzg,np	J/kg	soil energy (used to compute soil temperature)
SFCWATER_NLEV	nx,ny,np	#	number of snow (or surface water) levels
SFCWATER_MASS	nx,ny,nzs,np	Kg/m ²	surface water mass (snow + surface water)
SFCWATER_ENERGY	nx,ny,nzs,np	J/m ³	surface water energy (used to compute surface water temperature)
SFCWATER_DEPTH	nx,ny,nzs,np	m	surface water depth
USTAR	nx,ny,np	m/s	ustar
TSTAR	nx,ny,np	K	tstar
RSTAR	nx,ny,np	kg/kg	rstar
STOM_RESIST	nx,ny,np	s/m	leaf stomatal resistance
VEG_WATER	nx,ny,np	kg/m ²	vegetation moisture
VEG_TEMP	nx,ny,np	K	vegetation temperature
CAN_RVAP	nx,ny,np	kg/kg	canopy mixing ratio
CAN_TEMP	nx,ny,np	K	canopy temperature
GROUND_RSAT	nx,ny,np	kg/kg	saturation mixing ratio of the top soil/snow surface
GROUND_RVAP	nx,ny,np	kg/kg	Without snowcover, ground_rvap is the effective

			saturation mixing ratio of soil
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SURFACE CHARACTERISTICS (SIB only)			
(CAS = Canopy Air Space)			
RCO2P	nz,nx,ny	mass fraction	CO2 concentration (divide by 1.51724e-6 to get CO2 in ppm) (1.51e-6 = 44(g/mol) / (29g/mol) / 1.e6) where CO2 molar mass = 44 g/mol and Air molar mass = 29g/mol
SNOW1	nx,ny,np	kg/m ²	vegetation snow
SNOW2	nx,ny,np	kg/m ²	ground surface snow
CAPAC1	nx,ny,np	kg/m ²	vegetation liquid store
CAPAC2	nx,ny,np	kg/m ²	ground surface liquid store
PCO2AP	nx,ny,np	Pa	canopy air space CO2 concentration
CO2FLX	nx,ny,np	mol/m ² /s	surface CO2 flux (CAS to first atmospheric level) a.k.a. Net Ecosystem Exchange (NEE)
SFCSWA	nx,ny,np	fraction	surface albedo
UPLWRF	nx,ny,np	W/m ²	surface longwave upward radiation
ASSIMN	nx,ny,np	umol/m ² /s	uptake of CO2 by canopy plants
RESPG	nx,ny,np	umol/m ² /s	ground respiration flux
RSTFAC1	nx,ny,np	# (0->1)	leaf-surface humidity resistance stress
RSTFAC2	nx,ny,np	# (0->1)	soil moisture resistance stress
RSTFAC3	nx,ny,np	# (0->1)	temperature resistance stress
ECT	nx,ny,np	W/m ²	transpiration flux
ECI	nx,ny,np	W/m ²	canopy interception flux
EGI	nx,ny,np	W/m ²	ground interception flux
EGS	nx,ny,np	W/m ²	ground surface layer evaporation
HC	nx,ny,np	W/m ²	canopy (veg) sensible heat flux
HG	nx,ny,np	W/m ²	ground surface sensible heat flux
RA	nx,ny,np	s/m	CAS to lowest atmos layer aerodynamic resistance
RB	nx,ny,np	s/m	leaf surface to CAS aerodynamic resistance
RC	nx,ny,np	s/m	total canopy resistance

RD	nx,ny,np	s/m	ground to CAS aerodynamic resistance
ROFF	nx,ny,np	mm	water runoff (surface and subsurface)
GREEN	nx,ny,np	fraction	greenness fraction
APAR	nx,ny,np	fraction	absorbed fraction of photosynthetically active radiation
VENTMF	nx,ny,np	kg/m ² /s	ventilation mass flux
PCO2C	nx,ny,np	Pa	leaf chloroplast CO2 concentration
PCO2I	nx,ny,np	Pa	leaf internal CO2 concentration
PCO2S	nx,ny,np	Pa	leaf surface CO2 concentration
PCO2M	nx,ny,np	Pa	lowest atmospheric level CO2 concentration
EA	nx,ny,np	hPa	canopy water vapor pressure
EM	nx,ny,np	hPa	reference level vapor pressure
RHA	nx,ny,np	fraction	CAS relative humidity
RADVBC	nx,ny,np	W/m ²	visible direct radiation
RADVDC	nx,ny,np	W/m ²	visible diffuse radiation
RADNBC	nx,ny,np	W/m ²	near infrared direct radiation
RADNDC	nx,ny,np	W/m ²	near infrared diffuse radiation
PSY	nx,ny,np	hPa/deg	psychrometric constant

HYDROMETEOR MIXING RATIOS, NUMBER CONCENTRATION, ENERGY

SBM = Spectral Bin Model, mixr = mixing ratio

RCP	nx,ny,nz	kg/kg	cloud mixing ratio
RDP	nx,ny,nz	kg/kg	drizzle mixing ratio
RRP	nx,ny,nz	kg/kg	rain mixing ratio
RPP	nx,ny,nz	kg/kg	pristine ice mixing ratio
RSP	nx,ny,nz	kg/kg	snow mixing ratio
RAP	nx,ny,nz	kg/kg	aggregates mixing ratio
RGP	nx,ny,nz	kg/kg	graupel mixing ratio
RHP	nx,ny,nz	kg/kg	hail mixing ratio
RIPP	nx,ny,nz	kg/kg	plates mixing ratio (SBM only)
RICP	nx,ny,nz	kg/kg	columns mixing ratio (SBM only)
RIDP	nx,ny,nz	kg/kg	dendrite mixing ratio (SBM only)
CCP	nx,ny,nz	#/kg	cloud droplet number concentration

CDP	nx,ny,nz	#/kg	drizzle droplet number concentration
CRP	nx,ny,nz	#/kg	rain drop number concentration
CPP	nx,ny,nz	#/kg	pristine ice particle number concentration
CSP	nx,ny,nz	#/kg	snow particle number concentration
CAP	nx,ny,nz	#/kg	aggregates number concentration
CGP	nx,ny,nz	#/kg	graupel particle number concentration
CHP	nx,ny,nz	#/kg	hailstone number concentration
CIPP	nx,ny,nz	#/kg	plates number concentration (SBM only)
CICP	nx,ny,nz	#/kg	columns number concentration (SBM only)
CIDP	nx,ny,nz	#/kg	dendrites number concentration (SBM only)
Q2	nx,ny,nz	J/kg	rain internal energy
Q6	nx,ny,nz	J/kg	graupel internal energy
Q7	nx,ny,nz	J/kg	hail internal energy
FFCD	nx,ny,nz,nkr	kg/kg	cloud mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFIP	nx,ny,nz,nkr	kg/kg	plates mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFIC	nx,ny,nz,nkr	kg/kg	columns mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFID	nx,ny,nz,nkr	kg/kg	dendrites mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFSN	nx,ny,nz,nkr	kg/kg	aggregates mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFGL	nx,ny,nz,nkr	kg/kg	graupel mixing ratio distribution (SBM only) (d(mixr)/dln(r))
FFHL	nx,ny,nz,nkr	kg/kg	hail mixing ratio distribution (SBM only) (d(mixr)/dln(r))

**AEROSOLS MASS MIXING RATIOS AND
NUMBER CONCENTRATION**

CN1NP	nx,ny,nz	#/kg	CCN mode-1 number concentration
CN2NP	nx,ny,nz	#/kg	CCN mode-2 number concentration
CN3NP	nx,ny,nz	#/kg	CCN mode-3 number concentration
DUSTFRAC	nx,ny	fraction	Grid cell dust erodible fraction
MD1NP	nx,ny,nz	#/kg	sub-micron dust number concentration
MD2NP	nx,ny,nz	#/kg	super-micron dust number concentration
ABC1NP	nx,ny,nz	#/kg	Absorbing carbon (1% BC, 99% OC) number concentration
ABC2NP	nx,ny,nz	#/kg	Absorbing carbon (2% BC, 98% OC) number concentration
SALT_FILM_NP	nx,ny,nz	#/kg	sea-salt film drop number concentration
SALT_JET_NP	nx,ny,nz	#/kg	sea-salt jet drop number concentration
SALT_SPUM_NP	nx,ny,nz	#/kg	sea-salt spume drop number concentration
REGEN_AERO1_NP	nx,ny,nz	#/kg	sub-micron regenerated aerosol number concentration
REGEN_AERO2_NP	nx,ny,nz	#/kg	super-micron regenerated aerosol number concentration
CN1MP	nx,ny,nz	kg/kg	CCN mode-1 mass mixing ratio
CN2MP	nx,ny,nz	kg/kg	CCN mode-2 mass mixing ratio
CN3MP	nx,ny,nz	kg/kg	CCN mode-3 mass mixing ratio
MD1MP	nx,ny,nz	kg/kg	sub-micron dust mass mixing ratio
MD2MP	nx,ny,nz	kg/kg	super-micron dust mass mixing ratio
ABC1MP	nx,ny,nz	#/kg	Absorbing carbon (1% BC, 99% OC) mass mixing ratio
ABC2MP	nx,ny,nz	#/kg	Absorbing carbon (2% BC, 98% OC) mass mixing ratio

SALT_FILM_MP	nx,ny,nz	kg/kg	sea-salt film drop mass mixing ratio
SALT_JET_MP	nx,ny,nz	kg/kg	sea-salt jet drop mass mixing ratio
SALT_SPUM_MP	nx,ny,nz	kg/kg	sea-salt spume drop mass mixing ratio
REGEN_AERO1_MP	nx,ny,nz	kg/kg	sub-micron regenerated aerosol mass mixing ratio
REGEN_AERO2_MP	nx,ny,nz	kg/kg	super-micron regenerated aerosol mass mixing ratio
CIFNP	nx,ny,nz	#/kg	ice nuclei number concentration (Meyers/DeMott-limited schemes)
RIFNP	nx,ny,nz	kg/kg	ice nuclei mass concentration (SBM CNT scheme)
FNCN	nx,ny,nz,nkr	kg/kg	ccn mass mixing ratio distribution (SBM-only) (d(mixr)/dln(r))
FFIN	nx,ny,nz,nkr	kg/kg	ccn mass mixing ratio distribution (SBM only) (d(mixr)/dln(r))

AEROSOLS TRACKING			
IFNNUCP	nx,ny,nz	#/kg	ice nuclei already nucleated
IMMERCp	nx,ny,nz	#/kg	ice nuclei within cloud droplets
IMMERDP	nx,ny,nz	#/kg	ice nuclei within drizzle droplets
IMMERRP	nx,ny,nz	#/kg	ice nuclei within rain droplets
CNMCP	nx,ny,nz	kg/kg	total aerosol mass within cloud droplets
CNMDP	nx,ny,nz	kg/kg	total aerosol mass within drizzle
CNMRP	nx,ny,nz	kg/kg	total aerosol mass within rain
CNMPP	nx,ny,nz	kg/kg	total aerosol mass within pristine ice
CNMSP	nx,ny,nz	kg/kg	total aerosol mass within snow
CNMAP	nx,ny,nz	kg/kg	total aerosol mass within aggregates

CNMGP	nx,ny,nz	kg/kg	total aerosol mass within graupel
CNMHP	nx,ny,nz	kg/kg	total aerosol mass within hail
DNMCP	nx,ny,nz	kg/kg	total dust mass within cloud droplets
DNMDP	nx,ny,nz	kg/kg	total dust mass within drizzle
DNMRP	nx,ny,nz	kg/kg	total dust mass within rain
DNMPP	nx,ny,nz	kg/kg	total dust mass within pristine ice
DNMSP	nx,ny,nz	kg/kg	total dust mass within snow
DNMAP	nx,ny,nz	kg/kg	total dust mass within aggregates
DNMGP	nx,ny,nz	kg/kg	total dust mass within graupel
DNMHP	nx,ny,nz	kg/kg	total dust mass within hail
DINCP	nx,ny,nz	kg/kg	dust mass within cloud droplets via ice nucleation
DINDP	nx,ny,nz	kg/kg	dust mass within drizzle via ice nucleation
DINRP	nx,ny,nz	kg/kg	dust mass within rain via ice nucleation
DINPP	nx,ny,nz	kg/kg	dust mass within pristine ice via ice nucleation
DINSP	nx,ny,nz	kg/kg	dust mass within snow via ice nucleation
DINAP	nx,ny,nz	kg/kg	dust mass within aggregates via ice nucleation
DINGP	nx,ny,nz	kg/kg	dust mass within graupel via ice nucleation
DINHPP	nx,ny,nz	kg/kg	dust mass within hail via ice nucleation
SNMCP	nx,ny,nz	kg/kg	total soluble aerosol mass within cloud droplets
SNMDP	nx,ny,nz	kg/kg	total soluble aerosol mass within drizzle
SNMRP	nx,ny,nz	kg/kg	total soluble aerosol mass within rain
SNMPP	nx,ny,nz	kg/kg	total soluble aerosol mass within pristine ice
SNMSP	nx,ny,nz	kg/kg	total soluble aerosol mass within snow
SNMAP	nx,ny,nz	kg/kg	total soluble aerosol mass within aggregates
SNMGP	nx,ny,nz	kg/kg	total soluble aerosol mass within graupel

SNMHP	nx,ny,nz	kg/kg	total soluble aerosol mass within hail
RESOL_AERO1_MP	nx,ny,nz	kg/kg	sub-micron regenerated aerosol soluble mass mixing ratio
RESOL_AERO2_MP	nx,ny,nz	kg/kg	super-micron regenerated aerosol soluble mass mixing ratio

PRECIPITATION			
PCPVR	nx,ny,nz	kg/m²/s	rain precipitation rate (3D)
PCPVP	nx,ny,nz	kg/m²/s	pristine ice precipitation rate (3D)
PCPVS	nx,ny,nz	kg/m²/s	snow precipitation rate (3D)
PCPVA	nx,ny,nz	kg/m²/s	aggregates precipitation rate (3D)
PCPVG	nx,ny,nz	kg/m²/s	graupel precipitation rate (3D)
PCPVH	nx,ny,nz	kg/m²/s	hail precipitation rate (3D)
PCPVD	nx,ny,nz	kg/m²/s	drizzle precipitation rate (3D)
PCPVIP	nx,ny,nz	kg/m²/s	plates precipitation rate (3D; SBM only)
PCPVIC	nx,ny,nz	kg/m²/s	columns precipitation rate (3D; SBM only)
PCPVID	nx,ny,nz	kg/m²/s	dendrites precipitation rate (3D; SBM only)
PCPRR	nx,ny	kg/m²/s	surface rain precipitation rate
PCPRP	nx,ny	kg/m²/s	surface pristine ice precipitation rate
PCPRS	nx,ny	kg/m²/s	surface snow precipitation rate
PCPRA	nx,ny	kg/m²/s	surface aggregates precipitation rate
PCPRG	nx,ny	kg/m²/s	surface graupel precipitation rate
PCPRH	nx,ny	kg/m²/s	surface hail precipitation rate
PCPRD	nx,ny	kg/m²/s	surface drizzle precipitation rate
PCPRIP	nx,ny	kg/m²/s	surface plates precipitation rate (SBM only)
PCPRIC	nx,ny	kg/m²/s	surface columns precipitation rate (SBM only)
PCPRID	nx,ny	kg/m²/s	surface dendrites precipitation rate (SBM only)

ACCPR	nx,ny	kg/m ²	accumulated rain over the course of the simulation
ACCPP	nx,ny	kg/m ²	accumulated pristine ice over the course of the simulation
ACCPS	nx,ny	kg/m ²	accumulated snow over the course of the simulation
ACCPA	nx,ny	kg/m ²	accumulated aggregates over the course of the simulation
ACCPG	nx,ny	kg/m ²	accumulated graupel over the course of the simulation
ACCPH	nx,ny	kg/m ²	accumulated hail over the course of the simulation
ACCPD	nx,ny	kg/m ²	accumulated drizzle over the course of the simulation
ACCPID	nx,ny	kg/m ²	accumulated plates over the course of the simulation (SBM only)
ACCPIC	nx,ny	kg/m ²	accumulated columns over the course of the simulation (SBM only)
ACCPID	nx,ny	kg/m ²	accumulated dendrites over the course of the simulation (SBM only)
PCPG	nx,ny	kg/m ²	microphysics precipitation per timestep (for water, kg/m ² = mm), Used by LEAF/SIB surface models
QPCPG	nx,ny	J/m ²	microphysics precipitation energy per timestep, Used by LEAF surface model
DPCPG	nx,ny	m	microphysics precipitation depth per timestep, Used by LEAF surface model
ACCPDUST	nx,ny	kg/m ²	surface accumulated mass of aerosols identified as dust
ACCPAERO	nx,ny	kg/m ²	Total surface accumulated mass of aerosols
PCPRDUST	nx,ny	kg/m ² /s	surface accumulation rate of aerosols identified as dust
PCPRAERO	nx,ny	kg/m ² /s	Total surface accumulation rate of aerosols

RADIATION

FTHRD	nx,ny,nz	K/s	radiative heating rate
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FTHRDSW	nx,ny,nz	K/s	radiative heating rate SW component only
FTHRDLW	nx,ny,nz	K/s	radiative heating rate LW component only
BEXT	nx,ny,nz	km	visibility
SWUP	nx,ny,nz	W/m²	upwelling shortwave radiation
SWDN	nx,ny,nz	W/m²	downwelling shortwave radiation
LWUP	nx,ny,nz	W/m²	upwelling longwave radiation
LWDN	nx,ny,nz	W/m²	downwelling longwave radiation
RSHORT	nx,ny	W/m²	surface downwelling shortwave radiation
RLONG	nx,ny	W/m²	surface downwelling longwave radiation
RLONGUP	nx,ny	W/m²	surface upwelling longwave radiation
RLONTOP	nx,ny	W/m²	outgoing longwave at top of atmosphere
AODT	nx,ny	unitless	Aerosol optical depth in visible radiation band-3
ALBEDT	nx,ny	fraction	surface albedo
COSZ	nx,ny	unitless	cosine of the solar zenith angle

TURBULENCE AND FLUXES			
TKEP	nx,ny,nz	m²/s²	turbulent kinetic energy (from Mellor-Yamada and Deardorf schemes only)
HKH	nx,ny,nz	m²/s	horizontal eddy diffusivity coefficient for heat for scalar quantities (un-density weighted)
VKH	nx,ny,nz	m²/s	vertical eddy diffusivity coefficient for heat for scalar quantities (un-density weighted)
RHKM	nx,ny,nz	m²/s	horizontal eddy diffusivity coefficient for momentum
RVKM	nx,ny,nz	m²/s	vertical eddy diffusivity coefficient for momentum

RVKH	nx,ny,nz	m²/s	vertical eddy diffusivity coefficient for heat for scalar quantities
SFLUX_U	nx,ny	Pascals	surface U-momentum flux
SFLUX_V	nx,ny	Pascals	surface V-momentum flux
SFLUX_W	nx,ny	Pascals	surface W-momentum flux
SFLUX_T	nx,ny	(K*kg) / (m²*s)	surface temperature flux (multiply by Cp [~1004 J/(kg*K)] to get surface sensible heat flux in W/m²)
SFLUX_R	nx,ny	kg / (m²*s)	surface moisture flux (multiply by Lv [~2.5e6 J/kg] to get surface latent heat flux in W/m²)

CUMULUS PARAMETERIZATION FIELDS			
THSRC	nx,ny,nz	K/sec	convective parameterization heating rate
RTSRC	nx,ny,nz	kg/kg/sec	convective parameterization moistening rate
ACONPR	nx,ny	kg/m²	convective parameterization total accumulated precipitation
CONPRR	nx,ny	kg/m²/sec	convective parameterization precipitation rate
RCSRC	nx,ny,nz	kg/kg/sec	convective cloud water mixing ratio tendency (KF scheme only)
RRSRC	nx,ny,nz	kg/kg/sec	convective rain mixing ratio tendency (KF scheme only)
RPSRC	nx,ny,nz	kg/kg/sec	convective pristine ice mixing ratio tendency (KF scheme only)
RSSRC	nx,ny,nz	kg/kg/sec	convective snow mixing ratio tendency (KF scheme only)
WOAVG	nx,ny,nz	m/s	a running mean average of vertical velocity (KF scheme only)
WOAVGLT	nx,ny,nz	m/s	a running mean average of the horizontal components of the contravariant vertical velocity (KF scheme only)
NCA	nx,ny	m/s	integer counter keeping track of number of time steps that

			convective tendencies maintained (KF scheme only)
CONVGO	nx,ny	m/s	integer which keeps track if pre-convection checks satisfied (KF scheme only)

BUDGET VARIABLES for IMBUDGET >=1			
All budgets are accumulated (Total) values unless otherwise noted as being instantaneous values. Accumulated budgets are summed each timestep between analysis (A) output file writes and then reset. They are not reset for LITE or MEAN file outputs. "Total" budgets end with a "T". Instantaneous budgets have the same name but without the "T".			
WP_BUOY_THETA	nx,ny,nz	m/s	instantaneous vertical velocity contribution from Theta-V buoyancy computation
WP_BUOY_COND	nx,ny,nz	m/s	instantaneous vertical velocity contribution from condensate loading
WP_ADVDF	nx,ny,nz	m/s	instantaneous vertical velocity contribution by the combination of both advection and diffusion
LATHEATVAP	nx,ny,nz	d Θ or dT	instantaneous / single timestep (T or Θ) due to vapor diffusion and cloud & ice nucleation
LATHEATFRZ	nx,ny,nz	d Θ or dT	instantaneous / single timestep (T or Θ) due to collision-coalescence and melting routines
LATHEATVAPT	nx,ny,nz	d Θ or dT	change in (T or Θ) due to vapor diffusion and cloud & ice nucleation
LATHEATFRZT	nx,ny,nz	d Θ or dT	change in (T or Θ) due to collision-coalescence and melting routines
NUCCLDRT	nx,ny,nz	kg/kg	nucleation mass of cloud and drizzle
NUCCLDCT	nx,ny,nz	#/g	nucleation number of cloud and drizzle
NUCICERT	nx,ny,nz	kg/kg	total nucleation mass of pristine ice

NUCICECT	nx,ny,nz	#/g	total nucleation number of pristine ice
VAPLIQT	nx,ny,nz	kg/kg	vapor condensation summed for all liquid hydromet species
VAPICET	nx,ny,nz	kg/kg	vapor deposition summed for all ice hydromet species
EVAPLIQT	nx,ny,nz	kg/kg	evaporation summed for all liquid hydrometeor species
EVAPICET	nx,ny,nz	kg/kg	sublimation summed for all ice hydrometeor species
MELT2LIQTHERMT	nx,ny,nz	kg/kg	thermodynamic melting of ice species in melting routine that lead to mass transfers to liquid only species
MELT2RAINCOLT	nx,ny,nz	kg/kg	ice melting to rain due to collection of rain (rcy values; not "colxfers" adjusted; see mic_coll.f90)
MELTVAPT	nx,ny,nz	kg/kg	net melting computed from latent heating routines due to vapor and heat diffusion. Includes partial melting.
MELTCOLMELTT	nx,ny,nz	kg/kg	net melting computed from latent heating routines due to collision and melting schemes. Includes partial melting.
FREEZVAPT	nx,ny,nz	kg/kg	net freezing computed from latent heating routines due to vapor and heat diffusion. Includes partial freezing.
FREEZCOLMELTT	nx,ny,nz	kg/kg	net freezing computed from latent heating routines due to collision and melting schemes. Includes partial freezing.
FREEZICENUCT	nx,ny,nz	kg/kg	net freezing computed from latent heating routines due to ice nucleation.
CLD2RAINT	nx,ny,nz	kg/kg	cloud water transferred to rain via autoconversion + accretion

CLD2DRIZT	nx,ny,nz	kg/kg	cloud water transferred to drizzle via autoconversion + accretion
DRZ2RAINT	nx,ny,nz	kg/kg	drizzle water transferred to rain via autoconversion + accretion
RIMECLDT	nx,ny,nz	kg/kg	cloud water collected by all ice species (rcx values; see mic_coll.f90) (mass transfer from cloud)
RIMEDRZT	nx,ny,nz	kg/kg	drizzle water collected by all ice species (rcx values; see mic_coll.f90) (mass transfer from drizzle)
RIMERRAINT	nx,ny,nz	kg/kg	rain water collected by ice species (rcx values; see mic_coll.f90) (mass transfer from rain)
AGGRSELFPRIST	nx,ny,nz	kg/kg	mass transfer pristine ice to aggregates due to pristine ice self-collection
AGGRSELSNOWT	nx,ny,nz	kg/kg	mass transfer snow to aggregates due to snow self-collection
AGGRPSPRIST	nx,ny,nz	kg/kg	mass transfer of pristine ice to aggregates due to pristine ice / snow collisions
AGGRPSSNOWT	nx,ny,nz	kg/kg	mass transfer of snow to aggregates due to pristine ice / snow collisions
RAINBREAKUPT	nx,ny,nz	#/g	Raindrop number created by breakup from self-collection
CLDSIPHMT	nx,ny,nz	#/g	Pristine ice number created by Hallett-Mossop SIP with cloud drops
DRZSIPHMT	nx,ny,nz	#/g	Pristine ice number created by Hallett-Mossop SIP with drizzle drops
RAINSHEDT	nx,ny,nz	#/g	Raindrop number created by shedding of water from hail

BUDGET VARIABLES for IMBUDGET >=2

All budgets are accumulated (Total) values unless otherwise noted as being instantaneous values. Accumulated budgets are summed each timestep between analysis (A) output file writes and then reset. They are not reset for LITE or MEAN file outputs. "Total" budgets end with a "T". "Instantaneous" budgets for this set do not currently exist.

INUCHOMRT	nx,ny,nz	kg/kg	homogeneous droplet freezing ice nucleation mass
INUCHOMCT	nx,ny,nz	#/g	homogeneous droplet freezing ice nucleation number
INUCCONTRT	nx,ny,nz	Kg/kg	contact freezing ice nucleation mass
INUCCONTCT	nx,ny,nz	#/g	contact freezing ice nucleation number
INUCIFNRT	nx,ny,nz	kg/kg	condensation & immersion freezing ice nucleation mass from ice nuclei
INUCIFNCT	nx,ny,nz	#/g	condensation & immersion freezing ice nucleation number from ice nuclei
INUHAZRT	nx,ny,nz	kg/kg	haze droplet nucleation mass tied to aerosol concentration
INUHAZCT	nx,ny,nz	#/g	haze droplet nucleation number tied to aerosol concentration
VAPCLDT	nx,ny,nz	kg/kg	vapor condensation on cloud
VAPRAINT	nx,ny,nz	kg/kg	vapor condensation on rain
VAPPRIST	nx,ny,nz	kg/kg	vapor deposition on pristine ice
VAPSNOWT	nx,ny,nz	kg/kg	vapor deposition on snow
VAPAGGRT	nx,ny,nz	kg/kg	vapor deposition on aggregates
VAPGRAUT	nx,ny,nz	kg/kg	vapor deposition on graupel
VAPHAILT	nx,ny,nz	kg/kg	vapor deposition on hail
VAPDRIZT	nx,ny,nz	kg/kg	vapor condensation on drizzle
EVAPCLDT	nx,ny,nz	kg/kg	evaporation of cloud droplets
EVAPRAINT	nx,ny,nz	kg/kg	evaporation of rain
EVAPPRIST	nx,ny,nz	kg/kg	sublimation of pristine ice
EVAPSNOWT	nx,ny,nz	kg/kg	sublimation of snow
EVAPAGGRT	nx,ny,nz	kg/kg	sublimation of aggregates
EVAPGRAUT	nx,ny,nz	kg/kg	sublimation of graupel

EVAPHAILT	nx,ny,nz	kg/kg	sublimation of hail
EVAPDRIST	nx,ny,nz	kg/kg	evaporation of drizzle
MELTPRISTHMT	nx,ny,nz	kg/kg	mass transfer from pristine ice due to thermodynamic melting
MELTSNOWTHMT	nx,ny,nz	kg/kg	mass transfer from snow due to thermodynamic melting
MELTAGGRTHMT	nx,ny,nz	kg/kg	mass transfer from aggregates due to thermodynamic melting
MELTGRAUTHMT	nx,ny,nz	kg/kg	mass transfer from graupel due to thermodynamic melting
MELTHAILTHMT	nx,ny,nz	kg/kg	mass transfer from hail due to thermodynamic melting
MELTPRISCOLT	nx,ny,nz	kg/kg	mass transfer from pristine ice due to collisional melting
MELTSNOWCOLT	nx,ny,nz	kg/kg	mass transfer from snow due to collisional melting
MELTAGGRCOLT	nx,ny,nz	kg/kg	mass transfer from aggregates due to collisional melting
MELTGRAUCOLT	nx,ny,nz	kg/kg	mass transfer from graupel due to collisional melting
MELTHAILCOLT	nx,ny,nz	kg/kg	mass transfer from hail due to collisional melting
RIMECLDSNOWT	nx,ny,nz	kg/kg	mass transfer from cloud due to riming by snow
RIMECLDAGGRT	nx,ny,nz	kg/kg	mass transfer from cloud due to riming by aggregates
RIMECLDGRAUT	nx,ny,nz	kg/kg	mass transfer from cloud due to riming by graupel
RIMECLDHAILT	nx,ny,nz	kg/kg	mass transfer from cloud due to riming by hail
RIMEDRZSNOWT	nx,ny,nz	kg/kg	mass transfer from drizzle due to riming by snow
RIMEDRZAGGRT	nx,ny,nz	kg/kg	mass transfer from drizzle due to riming by aggregates
RIMEDRZGRAUT	nx,ny,nz	kg/kg	mass transfer from drizzle due to riming by graupel
RIMEDRZHAILT	nx,ny,nz	kg/kg	mass transfer from drizzle due to riming by hail
RIMERAINPRIST	nx,ny,nz	kg/kg	mass transfer from rain due to collisions with pristine ice
RIMERAINSNOWT	nx,ny,nz	kg/kg	mass transfer from rain due to collisions with snow
RIMERAINAGGRT	nx,ny,nz	kg/kg	mass transfer from rain due to collisions with aggregates

RIMERAINGRAUT	nx,ny,nz	kg/kg	mass transfer from rain due to collisions with graupel
RIMERAINHAILT	nx,ny,nz	kg/kg	mass transfer from rain due to collisions with hail

BUDGET VARIABLES for IMBUDGET >=3

All budgets are accumulated (Total) values unless otherwise noted as being instantaneous values. Accumulated budgets are summed each timestep between analysis (A) output file writes and then reset. They are not reset for LITE or MEAN file outputs. "Total" budgets end with a "T". "Instantaneous" budgets for this set do not currently exist.

DUST1CLDRT	nx,ny,nz	kg/kg	nucleation of cloud droplet mixing ratio from dust mode 1
DUST2CLDRT	nx,ny,nz	kg/kg	nucleation of cloud droplet mixing ratio from dust mode 2
DUST1DRZRT	nx,ny,nz	kg/kg	nucleation of drizzle droplet mixing ratio from dust mode 1
DUST2DRZRT	nx,ny,nz	kg/kg	nucleation of drizzle droplet mixing ratio from dust mode 2

KPP Ocean Mixed Layer Model Variables

Some of these KPP variables are required for history restart and must be output. Others are simply diagnostics. See RAMSIN flags and KPP code to control how many optional output variables are written to file.

KPP_OLD	nx,ny	index	Index to ID past value of Us,Vs
KPP_NEW	nx,ny	Index	Index to ID new value of Us,Vs
KPP_JERLOV	nx,ny	jerlov cat	Ocean optical clarity category
KPP_OCDEPTH	nx,ny	m	Ocean depth
KPP_HMIX	nx,ny	m	Mixed-layer depth
KPP_BOTTOMT	nx,ny	Celsius	Ocean bottom temperature
KPP_SREF	nx,ny	o/oo	Reference salinity
KPP_FREEZ_FLAG	nx,ny	fraction	Fraction of levels prevented from freezing
KPP_RESET_FLAG	nx,ny	Index	Flag to indicate isothermal column, reset T/S to climo
KPP_FLX_UST	nx,ny	N/m2	Sflux(1) zonal surface wind stress

KPP_FLX_VST	nx,ny	N/m2	Sflux(2) meridional surface wind stress
KPP_FLX_NSW	nx,ny	W/m2	Sflux(3) Net surface shortwave radiation
KPP_FLX_NLW	nx,ny	W/m2	Sflux(4) non-shortwave radiation (lwdn-lwup-sensible-latent)
KPP_FLX_ICE	nx,ny	Not used	Sflux(5) melting of sea ice
KPP_FLX_PCP	nx,ny	mm/sec	Sflux(6) net freshwater (precip-evaporation)
KPP_SWDK_OPT	nx,ny,nkppz	fraction	Solar shortwave flux fraction on DM depths
KPP_wB	nx,ny,nkppz	m2/s3	w'B' total kinematic buoyancy flux
KPP_wU	nx,ny,nkppz	m2/s2	w'U' turbulent zonal velocity flux
KPP_wV	nx,ny,nkppz	m2/s2	w'V' turbulent meridional velocity flux
KPP_wXt	nx,ny,nkppz	C * m/s	w'T' turbulent temperature flux
KPP_wXs	nx,ny,nkppz	o/oo * 1/s	w'S' turbulent salinity flux
KPP_wXNTt	nx,ny,nkppz	C * m/s	w'T'(NT) non-turbulent temperature flux
KPP_SWFRAC	nx,ny,nkppz	fraction	Solar shortwave flux fraction on ZM depths
KPP_TINC_FCORR	nx,ny,nkppz	K	Temperature increment from flux correction with depth
KPP_SINC_CORR	nx,ny,nkppz	o/oo	Salinity increment from flux correction with depth
KPP_SAL_CLIM	nx,ny,nkppz	o/oo	3D salinity climatology (can be time updated)
KPP_OCNT_CLIM	nx,ny,nkppz	Celsius	3D temperature climatology (can be time updated)
KPP_BUOY	nx,ny,nkppz	m/s2	buoyancy
KPP_RHO	nx,ny,nkppz	kg/m3	Ocean density
KPP_CP	nx,ny,nkppz	J/kg/K	Ocean specific heat capacity
KPP_U	nx,ny,nkppz	m/s	Latest value of ocean U current
KPP_V	nx,ny,nkppz	m/s	Latest value of ocean V current
KPP_U_init	nx,ny,nkppz	m/s	Initial value of ocean U current
KPP_V_init	nx,ny,nkppz	m/s	Initial value of ocean V current

KPP_US0	nx,ny,nkppz	m/s	Contains old/new U current
KPP_VS0	nx,ny,nkppz	m/s	Contains old/new V current
KPP_US1	nx,ny,nkppz	m/s	Contains old/new U current
KPP_VS1	nx,ny,nkppz	m/s	Contains old/new V current
KPP_X_T	nx,ny,nkppz	Celsius	Latest ocean temperature
KPP_X_S	nx,ny,nkppz	o/oo	Latest ocean salinity (+sref)
KPP_XS_T0	nx,ny,nkppz	Celsius	Contains old/new temperature
KPP_XS_T1	nx,ny,nkppz	Celsius	Contains old/new temperature
KPP_XS_S0	nx,ny,nkppz	o/oo	Contains old/new salinity
KPP_XS_S1	nx,ny,nkppz	o/oo	Contains old/new salinity

CUSTOM TRACER VARIABLES			
TRACERP001	nx,ny,nz	mixing ratio units needed	customizable by the user
TRACERP002, etc...			